

It's like two lectures a week. Okay. Oh, that's right. So good afternoon, everyone.

My name is Carrie Coyle. I'm one of the pulmonary critical care attendings at the hospital, and I have the pleasure of working with you guys pretty much all throughout your careers So in the modules now with the first years. I'm here with you guys for the pulmonary module, and I get students and I mentor and whatever. So it's certainly super fun for me to be a part of your active learning for those of you who don't know me The reason why I find it really enjoyable is because I literally sat in your seats. I was a med student here, and I'm a graduate of the class of 2006. So it's super fun for me to come back and sort of give back To what, you know, was given to me.

I loved my time at NYMC, and I still love it now. So we have a very huge Sort of topic and introduction to pulmonary disease.

And then at the end there'll be a section on pulmonary preventive care Some of it, as you see as we go through, I'll be able to just sort of hastily go through the slides because you've had some of this with Dr.

Moshman, and it will be repetitive, but I wanted to create slides so that you guys could reference it if you found that helpful for your studying and future. So we'll get started We have a number of objectives that we have to get through today. We're going to learn how you should take a medical history For your patients who have respiratory disease, we're going to try to understand some of the pathophysiology and the presenting symptoms.

The main ones we'll focus on today are dyspnea, cough, chest pain, and hemoptysis. We're going to know the technique and interpretation Of findings on the physical exam, lung and thorax. This is the section we'll sort of hastily go through.

We're gonna figure out what are some invasive and non-invasive pulmonary tests that we can use to help us diagnose. We're going to go through some risky behaviors for lung disease that'll include smoking Vaping of nicotine.

We'll also touch base on marijuana cannabis today, and some other lifestyle modifications. Then we'll review, we'll go into the preventive section for Vaccines and then some other preventable diseases as well.

Most... one of the most important for prevention right now that we're seeing is for LT, latent TB So let's get started.

Throughout this, we'll sort of be referencing back to these cases. Our first case is a 23-year-old college student who was previously healthy. They're presenting acutely with Fever for 4 days, they have some body aches, they're fatigued, and they have a non-productive cough Our second case is a 58-year-old prior smoker. They quit 10 years ago, and they're coming in more subacute 3 weeks of cough with scarce yellow sputum, severe shortness of breath and weakness, and our third case is that of a 13-year-old male.

He has a known history of severe eczema, some environmental allergies, and he's coming in with progressive Exertional shortness of breath, wheeze, and a cough. And when he rests, the symptoms improve. So the things that you have to think about when you're looking at your patients and all patients in general is you want to observe the patient. So when you approach a patient Oftentimes you'll be seeing them walking into either an exam room or if you're in a hospital setting, you'll be coming in from the hospital corridor into their room, and you can see if they don't know that you're there before you make yourself known. You can see from the doorway whether they are in a comfortable state, they're comfortable, casual, on their phone, you know, just waiting for you, or if they look like they're uncomfortable, that's important for you to already be assessing as you approach them You have to go over their entire Hpi, and most importantly, for pulmonary patients are their social history, their occupational history, and

their travel history, because those things can affect their lung health. Always listen very carefully and listen to their questions and answer their questions.

You want to be able to identify important associated symptoms to their HPI, and you are always focused on what their chief complaint is. So we're going to talk about shortness of breath. This is a very big topic. A lot of patients come in to see me regularly for this sensation of shortness of breath. There's two different presentations that you can have, or even three. There's the acute presentation, where they've only had this symptom for a very short amount of time, a chronic where it's been there for more than 4 weeks, they can have acute on chronic so that they always have a little bit of shortness of breath in the background, but then they then have another acute presentation on top of it. So some of the things that you may see in patients who have shortness of breath or those who have chronic lung conditions, like if they're a chronic bronchitis, a COPD, an asthma patient, if they have heart conditions like CHF or arrhythmias, sometimes patients who have chronic atrial fibrillation will describe being short of breath. If they're anemic and anxious, they could also feel short. Okay, so the pathophysiology of dyspnea is pretty involved.

There are different signaling pathways that will contribute to the sensation of being short of breath. So the reason why it's pretty complex because it involves different sensors. And acid base status and your partial pressure of oxygen, and it has chemoreceptors, mechanical receptors, and any disturbance can result in the sensation of dyspnea.

So if someone is anxious and hyperventilating, that's going to affect their carbon dioxide, in which case that's going to trigger one of these receptors, right, your chemoreceptor, and then it's going to tell your body to do something.

So it's all looped together. The important thing for you to start thinking about when someone says, oh, I've been short of breath for 1 day, is you're going to want to be thinking about the diagnoses that can happen in the acute presentation, and some of these could be pneumothorax, which is a collapsed lung. Pulmonary embolism?

Did they accidentally inhale a foreign body? Could they be having laryngeal edema? Meaning like they ate a nut and they didn't know that they were allergic, so they could be having an anaphylactic reaction? Or are they having pulmonary edema? And you can see that in the more subacute are your patients who have perhaps chronic underlying pulmonary diseases or are they coming in with a pneumonia that's been brewing over a couple of days. And then as you move into your more chronic causes of shortness of breath.

So you have to remember in your HPI, which we teach you as a MS1, is, well, what makes things better, what makes things worse. So you always want to be asking those questions with their presentation. So if they tell you that mainly they're short of breath with exertion, with doing things. That's very commonly seen in chronic pulmonary and cardiac conditions. There's another term where called PND where if the patient is laying down and they awaken suddenly after falling asleep, and then they have to get in an upright position and sometimes even they say that they need to run and open up a window and get some fresh air. That's very commonly seen in patients with underlying congestive heart failure. And then there's different positions that might help you figure out what type of condition they might have that's affecting them to cause dyspnea.

And that's certain things of Bend apnea, like when they bend over, it's very common for patients to say, when I bend over to tie my shoes, I lose my breath, and the reason being is that when they bend over, if they have a very large, whatever abdominal girth when they bent over, all of that is now squishing their lungs because their diaphragm is going to be pushed up and they're going to get short of breath. It's also seen in CHF. Orthopnea when they're laying flat, and if they get up, and that's laying flat awake.

When they get up there, it feels better. That's commonly seen in CHF. Again, obesity because the abdominal structures are coming up and squishing on the diaphragms and then inspiratory muscle weakness, which could be seen in some of your neuromuscular disorders. That's another common complaint, that when they lay down, they feel short of breath. So we're going to get to pulmonary testing at some point, but in somebody who doesn't appear to be, you know, very obese or have congestive heart failure, but they're describing that Orthopnea complaint, you would do breathing tests on your patient in an upright and supine position to help diagnose an underlying neuromuscular disorder, in which case, when they lay flat, you would see that their pulmonary function is worse. Okay, this is another major complaint that you'll come across in your clinics.

Patients who are complaining of cough. The cough reflex is also very, very, very in-depth. Essentially it's a pathway that has to get stimulated by something.

That stimulation then goes on to the receptors, stimulates the receptors, and then it goes on to the mechanical receptors, which are essentially the things that are going to move the muscles to then make the cough.

So again, you will have your afferent and efferent pathways. A cough is really, really strong, can be really strong.

It can create positive intrathoracic pressures as high as 300 millimeters of mercury. So we've had patients before who have actually cracked ribs, have rib fractures. Pulled muscles, having some muscle strain from coughing so much. And with everything, there's a timeline for a presentation.

So you have acute, subacute, and chronic. And patients who come in acutely with a cough, the most common thing that we'll see is someone who has a URI. And then in those who are coming in with a chronic cough, those things are more likely to be your patient who has untreated GERD or they have upper airway cough syndrome, or they have undiagnosed asthma or poorly controlled asthma, those are your common things for the chronic. The subacute in between is usually something that just hasn't fully been diagnosed yet, meaning that they're not sick with a URI, but perhaps they have allergic rhinitis or something like that, and they have a bunch of post-nasal drip that's making them cough. Pneumonic is listed for you guys here to try to help you remember different types of things that can make you cough. Very commonly on test questions, there could be a medication that you missed on talking to them, and a very common one is a different ACE inhibitors. And that could be something that could make them cough. And they, for all we know, could have been on this ACE inhibitor for two years already. The timing of when they started it doesn't matter. It could still make you cough at any point. Patients might only complain, or mostly complain of coughing at night.

This is very common in patients who have postnasal drip because it's positional. So when they lay down, it's going to pool in the back of their throat, and any of you guys who have had a really bad upper respiratory infection, I'm sure, can attest to some sleepless nights when you've been pretty sick, because you're just coughing your guts out because of that overstimulation of the receptors in the back of your throat. GERD is another very, very common presentation of patients laying supine.

What happens is the muscles relax and then the acid will come up and it will hit the upper airway, and it will cause them to cough. You have to talk sputum with your patients. So is the cough productive or non-productive? So if a cough is productive, you want to know all the gory details about it. And patients are very excited to talk to you about it. So they'll show you pictures. They'll bring samples in for you, even if you're not trying to solicit them, they want to show you. But it's important for you to look at it and to talk about it because it can help you figure out what the heck is going on with your patient.

So in patients who have a ton of mucus, it could be something very worrisome for something acute like a lung abscess. Did they have a really bad lung infection

that's now turned into an abscess or a necrotizing pneumonia Or could it be something that they have a more chronic lung condition called bronchiectasis where they're just constantly creating mucus? There's different consistencies to the mucus, that's important. So if it's very mucoid, that could be like a chronic bronchitis. If it looks like there's pus in the mucus, that's most likely an acute bacterial infection The color matters.

So usually your yellow green mucus is a bacterial infection. If it's pink and frothy, that's very common for patients with CHF to have pulmonary edema.

A rusty color is commonly seen in pneumococcal pneumonia, and then you have red currant jelly, which is seen in Klebsiella pneumonia. Patients will describe their mucus as being blood-tinged, and that can sometimes be a sign of tuberculosis.

And then there's anchovy sauce. And these are fun little terms that you'll come across in your reading and your testing questions And if you see these terms, this is sort of what it's pointing to. So on the slide, these are examples of what red currant jelly would be like literally looks like the jelly that you'd put on your PB&J.

And then this is an example of like anchovy paste, anchovy sauce that could be seen in a ruptured liver abscess You want to, yes it is.

So you want to also listen if they tell you about like it smells. So bronchiectasis has very copious, smelly sputum, and so do lung abscesses or anaerobic bacterial infections.

So perhaps your patient, you know, accidentally aspirated those type of lung infections are often very foul smelling Chest pain, you'll see a lot. It's either going to be something emergent Meaning cardiac related or it could be something less emergent but still super important, and it could be something involving your pulmonary system.

So chest pain doesn't always mean cardiac, okay? You want to be able to talk to your patient with your in-depth HPI about well you know what possibly could be happening with this chest pain.

And you can figure that out by what they describe, meaning that if they talk about it in a certain part It could correlate with, you know, different disease things.

So a lot of patients could describe that They have a chest pain that is much worse when they take in a deep breath.

That's something that suggests that it could have inflamed your pleura. So that could be something like a pulmonary embolism, where they had a pulmonary embolism causing a pulmonary infarct And now it's affected the pleural lining and it hurts every time they take big breaths in. hemoptysis.

This is when this means literally coughing out blood. So Coughing out blood tinges on mucus is different than coughing out physical blood, and coughing out physical blood can be extremely life-threatening, so the amount matters And What you need to know is that severe life-threatening hemoptysis is having about 150 milliliters in a 24-hour period, which is, just so you know, this one right here is 100 milliliters.

That's a lot of blood Right?

So a bag of blood has 250 milliliters in it, and a bag of blood, I'm sure you've all seen in some capacity, and that's a lot. So 150, right, is Pretty significant or it's also considered life-threatening if they're doing more than 100 milliliters in an hour And it's important for you to find this out because if you don't know how much they're coughing out.

The potential for them to die is extremely high. So The reason why it's important for you to figure out what exactly it looks like is because it can narrow down where it's coming from. So if it's coming from The pulmonary arterial system versus the bronchial arterial systems, so the majority are going to be coming from your PA circulation. These are a low pressure system. These are typically considered the non-life-threatening ones, but in some cases they can come from the bronchial arteries So Those are much less common, only about 2% of the They will cause Up to 90% of the ones that kill patients, but they're much less common. So there's a number of different reasons why people could be coughing out blood. And We see it most commonly in like a lung infection, a bronchitis. Usually it's going to be blood mixed with the sputum and not pure liquid blood. That can happen in pneumonia, tuberculosis, bronchiectasis, CF You can also cough out blood if you have a PE Obviously, if you have lung cancer, but other causes that you want to think about are if they, you know, are recreational drug use patients.

Sometimes those who are inhaling, like, crack cocaine or something, they can wind up having hemoptysis Those who are over, their blood is too thin, maybe they're taking accidentally too much blood thinners That can happen and foreign bodies, it's common to see this in the younger population who accidentally inhale something into their airway.

And then it's also important for people to figure out, is it coming from the coming from the lungs, or is it coming from upper? So oftentimes, patients will have epistaxis And then they'll think that it's hemoptysis, but it turns out that it's not coming actually from the pulmonary vasculature. These are the different, the most common types of lung cancers that would be presenting with hemoptysis. And AV malformations, mitral stenosis, and pulmonary hypertension are some of the most common cardiovascular reasons to have hemoptysis. Your college and vascular disorders are These are the ones listed are the ones that could potentially present with hemoptysis as a vasculitis flare. Okay, this is another new mnemonic. Again, I put a lot of slides together for you guys to come back and use them as study aids. So we're not going to go through every single thing on every slide. But battle camp is the one that you should try to have in your head. So case one, I hope you could all recognize, was an acute presentation.

Case two was a subacute presentation, and case three is a chronic presentation. So this is the area, this section of the slides is the one that I'm really going to sort of speed through because you have already had this with some of your other professors But it's very important for you to know how to do all of these techniques. Meaning that you have to look at your patient first because it gives a lot of hidden gems that you can use. So if you see, for example, example, that your patient has a you know Malor rash on their face, you might be thinking of an underlying rheumatologic disorder that could be leading to a lung problem. You have to know how to palpate and where to palpate on your patients, how to percuss, where to percuss And certainly auscultation. And these are all skills that you'll continue to hone in throughout your career. You started doing them as a MS1, and you're going to continue growing on them, you know, throughout your career. So like I said, these are there for your reference.

You've reviewed all of these techniques, I know, and you will continue to review these techniques. But it's important for you to be able to decipher what the pathology that can happen, that can result in some of these changes. Percussion, I would say the one thing that's most important is to really learn how to do percussion, you should be practicing on yourself because you can percuss all over your body and you'll learn the technique If you do it right on yourself, you'll definitely get it right on your patients. These are the different zones that you should be auscultating. And there's a pattern that you have to follow to auscultate. There are different breath sounds. Again, you should be practicing on yourself and friends and family members as much as they'll let you to learn what normal breath sounds sound like so that when you hear something abnormal, you'll know it's abnormal and then you can try to figure out what

abnormal sound it is and why it's happening. Okay.

As you know, there's all different types of lung sounds that you can hear. And once you learn them, then you'll be able to try to figure out what possible disease process could be happening. Decreased breath sounds can happen for a number of reasons. It could be that their lung is not inflated, and that's why it's low. It could be because they have emphysema. Maybe there's a big pleural effusion there. Vocal sounds again.

Some really fun ones that you'll get to experience throughout your career are bronchophony, egophony, and whispered pectoriloquy. These, obviously, there has to be a pathology present for you to be able to hear them. So it will come during your rotations, but again, you have to know what is normal first to pick up the abnormal. So we'll now go to reference back to the case studies, and we're going to see that our case 1 patient has an O2 sat of 96%. They're a little bit tachypneic. They have a flushed appearance, they're febrile, their lungs are clear and equal bilaterally. Our second patient is tachypneic.

They have crackles and they are hypoxic. Their saturation is only 85%. Our third patient has good O2 sat, mildly elevated respiratory rate of 16, good air entry, and they have wheezes. So now we're going to say, well, how are we going to diagnose all of these 3 patients that are in front of us? And these are some different ideas that we have to try to figure things out. There's different pulmonary testing. This is not inclusive, but these are the big ones that you'll be coming across. So there's PFTs, which are pulmonary function tests. We have radiographic imaging, which is your x-rays, your CAT scans.

MRI is not commonly used on the lungs. It's not a great test for the lungs, but certainly it's great for other organs. A biopsy if we need to get tissue of the lung. Bronchoscopy, which is a procedure done to help get biopsy or get samples, and then thoracentesis, which is a procedure that we do to sample fluid around the lung. So there are a number of reasons why you might want to get a PFT. This is for our patients who are coughing, wheezing, short of breath. Maybe they have exertional cough or chest pain. You want to find out if you give them a bronchodilator such as albuterol or ipratropium or levalbuterol. Do they respond to it, meaning does their lung function improve after it? Maybe they're coming to you because we want to rule out some type of occupational exposure, and that's important to employment for them. They, like, I have a number of firefighters, police officers who come to me for routine testing to make sure that their lungs stay healthy at work. We want to look for people who might have to undergo surgery, and we need to know if it's safe for them to undergo surgery or if we have to talk to the anesthesiologist to say, look, I really don't think that they should undergo general anesthesia based on X, Y, and Z. If we just need to know, do they have respiratory impairment? So if we think that maybe somebody has a neuromuscular problem, we can measure the strength of their lung function. And then, obviously, the most common one is that you see a patient, you diagnose them with something, you give them medicine to try to make them feel better, and then you want to see, well, how much better is their lung function so that you now have their new baseline. Bronchoscopy, this is what we call an ambulatory procedure meaning that patients will come in sort of as if they're having a colonoscopy. They go, they get anesthesia, usually light anesthesia, and they go home the same day. But we can do these anywhere in the hospital. We do them bedside in the medical ICU as well, so it just depends on what your indication is.

So, the most common indication for bronchoscopy is, do you think something is in the airway and shouldn't be? Is there an infection that we need to get a sample of? Is there something abnormal like a mass that we need to biopsy because we're concerned that maybe there's a malignancy there? So there's usually reasons that we have to go in to get a sample. This is a reason for hemoptysis. So if somebody is coming in and it's not just little blood tinges, it's the real thing, we need to go in to find out where's the bleeding coming from, because if we figure out that it's coming from something in the airway, we can zap it and sort of cauterize it, or we could take out the thing in the airway.

Or if it's from a larger process in the lung we put a blocker in to block that region off. Sometimes we can put glue in there, so there's all different things that we can do to stop the bleeding, because if the bleeding continues and gets throughout the airway, they die because essentially their airways become full of blood, they can't ventilate, they can't oxygenate, and they drown in blood. A big common reason is for malignancies, right, so we'll go in there. This guy will not only biopsy and send for pathology, but we'll take it out to reopen up that blocked airway. Foreign bodies are very common, more common in the younger patients, but they happen in adults as well. And these are all examples of a tooth that accidentally got knocked off and put in. In the airway pneumonia, if we need to know to get a very specific type of sample.

Sometimes patients are not able to expectorate the sputum, so we can't test it that way. Thoracentesis is when we go in with a needle from the outside through the rib, and we collect the fluid.

So this is a chest X-ray here, and all of this area here is the pleural effusion. This is the pleural effusion here, this little guy here is the lung, and this is the diaphragm. So the needle would come through into here and you'd sample the fluid. These are some different examples of what fluid can come out looking like. So sometimes, you know, this could be from a cancer or from an infection. This is pretty you know, clear, normal looking, but there are certain things called urinothoraxes, and then this would be like a chylothorax. And you take this fluid and you send it for sampling, and it's important for you to say, well, it's simple fluid, it's transudative, or it's complicated fluid, it's exudative. And the reason why that's important is because there are different diagnoses and treatments that would be done. So, you guys will become very familiar with how to diagnose these criteria.

You have to have lab work as well to compare and light's criteria. And you're looking at protein and LDH. And that'll help diagnose your criteria. Some of the easier or more modern ways of doing it is you want to look at the plural cholesterol. And if that's super elevated, then that tells you it's an exudative. But these are things that you will certainly become quite experienced with. Biopsies, these can be done from the outside in or from the inside.

So the first picture here is done by interventional radiology. This is called a CT-guided lung biopsy. And essentially they're taking a needle and going through the chest wall and into the lesion. This is a VATS.

So this is done by our cardiothoracic surgeons where they put in ports and they actually can go in that way. When this happens, the patient is completely under general anesthesia. In this case, the patient can be wide awake but is often giving some type of sedative, and then a bronchoscopy. Patients are also given some type of sedative, but they can be done awake in very risky patients. You will learn about all of these types of different radiology studies from Dr.

Racklin. Okay, so then we did some pulmonary testing on our patients. So on the first patient, we did a spirometry and it came back normal. On our second patient, we got a CAT scan. Well, we did a spirometry that was moderately restricted, and we got a CAT scan after that because we're figuring out do they have some type of interstitial lung disease? And it's showing that they have ground glass opacities, honeycombing, they have traction bronchiectasis. Our 3rd patient has moderate obstructive lung disease, and when we gave him a bronchodilator, it got better, and his X-ray showed that there was hyperinflation of his airways. So our first one, we thought that they had acute influenza.

Our second one, we think that they have idiopathic pulmonary fibrosis. And our third one is poorly controlled asthma. Okay. So this is a section where we're going to talk about smoking, vaping, and marijuana. So smoking, regrettably, is the number one cause of preventable disease and death worldwide. It claims more than 480,000 lives alone in the United States every year. That's nearly 1 in 5

causes of death in the United States.

It is super, super costly. So in the US alone, it costs at least \$600 billion each year, and that comes from lost work Productivity. So if they have COPD and they're having COPD exacerbations and they're not able to come into work or they've developed a pneumonia And then there's also on top of that, the lost productivity at work results in consumption of healthcare.

So there's about \$260 billion in direct health care as a direct result of smoking. In 2022, about 12 of every 100 US adults over 18 years old was considered an active smoker So what's important for you to recognize is that the majority of people who are smokers is because that they've been smoking at a young age And almost 94% by the age of 21.

So our job as providers is to be talking to all of our patients, even our young patients, about Not starting to smoke, because If you start to smoke, it's highly likely you will continue smoking. So some of the things that are, you know, scary that you could particularly talk to your patients about is to say, well, if you're a smoker, you have a 50% chance that you're going to die way younger than all your friends who don't smoke And definitely about 10 years before your friends who've never smoked.

So I think that's kind of scary and you should use that information. So It's still higher in certain types of demographics in the United States Men smoke more than women still.

If you come from a lower education, if you are uninsured or underinsured, if you're disabled, if you have mental illness, certain parts of the Like where you live in the United States matters.

So in those areas, the hope would be that education and prevention would be really high to try to intervene on those patients. So cigarettes have a ton of stuff in them besides nicotine.

And you get the thing that keeps you hooked on cigarettes is the nicotine. However, you're smoking all of these chemicals and toxic things which there's more than 7,000 other things than nicotine and cigarettes that they found and 69 can cause cancer. So For me as a pulmonologist, what's Awful to know is that 90% of all lung cancers are a result of smoking And that more women will die from lung cancer than from breast cancer every year. But I feel like a lot of the media is focuses on breast cancer for women Maybe women don't even know that they're more likely to die from lung cancer than breast cancer, so I think there's a lot that we could do in the United States in terms of education. Smoking causes about 80% of all deaths from COPD Cigarette smoking increases the risk for death from every single cause in men and women That the risk of dying from cigarette smoking has increased over the last 50 years and continues to increase. And you're more likely to have heart disease, you're more likely to have a stroke as a result of smoking It essentially affects every single part of your body.

So What would be ideal is if is that if nobody smoked one of every three cancer deaths in the United States wouldn't even happen. So when talking to your patients, let them know that it's not just they're not just hurting their lungs, they're hurting or essentially everywhere, from head to toe inside of their body is affected by what they're doing. There are a number of different tobacco products that you guys should learn about Because your patients are using them, and if you are educated, you'll be able to help them with their addiction. and understand where they're coming from. So the most common one, obviously, is cigarettes, and it depends on what age group that you're talking about. But a lot of patients will be using e-cigarettes or vapes Hook has made like a really big presentation within probably the last 10 years.

A lot of people are doing hookah parties. And then very recently, pouches, zin pouches are coming on on the scene over the last, probably, like, 5 years, huge

within the teen population. So I get the question all the time of, well, who's vaping, what's in a vape?

Vapes are safe, right? So vaping is literally exactly what it says. You're inhaling a vapor product Of something that gets heated up inside the pen and then you're breathing in those chemicals. What you need to know is that it's again, not just nicotine that you're breathing in the vape. There's also stuff in there that causes cancer, and there's stuff in there that causes chronic lung disease And people need to know that it's not safe. There's a huge push to say vaping is safer than smoking Is it safer? It depends on how you want to say that. Does it cause less harm? Yes. But it doesn't cause no harm. It's been really popular in the younger aged people because it's easier to hide and unfortunately, media has really attacked our children so that now kids are quite hooked on vaping. So what I want you to see is that in children It starts really, really young.

So when they pool students starting in middle school, middle school's young guys, right? So middle schoolers is like sixth grade. That's 11 years old. 4% of those kids have already used some type of tobacco product About 10% of high school students have done that as well. And I can share a personal experiences that I have a 10-year-old and a 14-year-old.

And I always go to, they do a health fair for the younger school. And this year I went and I talked to them about vaping And as sort of a joke, I said, hey, raise your hand for any of you guys in the classroom who like to vape Do you understand that 4th and 5th graders were raising their hand? I was horrified because I didn't. I didn't think that that was going to be the response. But I had to be able to react to them. And it turns out they're getting it from their parents The parents leave it around. They're flavored, they're shiny, they're whatever. And They're using their parents' vapes and they're vaping in fourth and fifth grade already. And they didn't understand that it was dangerous, and they didn't understand that it was bad, and they didn't understand that it was addictive because their parents aren't having these conversations with them. So I'm sure that was a very interesting dinner discussion they had at home that night, but just know that it's very common It's because of the tobacco industry.

They're actually targeting our children. So they get lured in by these things. My 14-year-old helped me with some of these slides, I have to admit. I have no idea what this guy is, but apparently it's big in the media But these things here are vapes, and they're interactive, where they can, like, play video games on them And then these things over here look like candy to me. They look like push pops or whatever I remember from being a kid. So you can see why they look at these and they think that they're not harmful. So they start using them. This is a slide that I showed the fourth and fifth graders to play. Can we find which one is not a vape All right Does anybody see any like raise your hand if you think that there's something on here that is not a vape. The Sharpie.

Guess what? It's a vape. I mean, this is crazy, right? So this is where also as parents, we can certainly be fooled and tricked. You know, if you think this is, you know, this is like a little like USB drive Why would you think that that was like weird? You'd be like, oh, my kid's doing their homework, it's so great, right? And you don't even pay attention to that. You... they're like, oh, can I have this new hoodie sweatshirt or a t-shirt, whatever? And you're like, okay, sure. Well, inside of that, they're vaping, so those little strings they're putting in their mouth, and they're vaping at school So this is really scary, okay, and I think it's important for you guys to have an idea of what these things look like because you'll be doing pediatric rotations and all these things and you might actually catch something that maybe a parent didn't, and that'll be a good intervention and prevention for you So they do it as we know to be cool, or they're curious, or it's on social media. They clearly don't understand that it's harmful. And I had that personal experience myself And then what happens is they continue to vape because they get addicted So we know that if they start vaping these like yummy nicotine products, they're more likely than 4 times more likely to go on to smoking regular cigarettes And then they

will likely, there's a higher likelihood of them going on to get addicted to other drugs of abuse. So this is a slide for my daughter again. I don't know, she thinks it's cool. But what's very important is that the kids who have vaped The ones who have ever vaped, 43% of them are still currently vaping That's almost half of kids who tried it are still vaping. Because they are so addictive And when you talk, when they've done these surveys and talking to kids in middle school and high school, almost 70% of them have actually tried to quit and they can't Pouches are really big, like I said, over the last five years. ZIN is the number one sold brand of nicotine pouches.

The reason why kids like to use these things is because they can look like gum. So there's like gum tapes or gum containers, and they can sort Hide that from their parents, they can put it in their pocket easily, and they have access to being able to do something that doesn't have smoke.

They don't have to inhale or exhale. It's super easy for them to hide. They're mostly flavored. So again, they are hooking the kids based on some type of yummy flavor that they go after. So Between 22 and 25, the youth in young adults has quadrupled There, the second most popular type of tobacco product after vaping in the young kids.

There is such a thing as zinfluencers. So they're essentially telling them that if they use it, they'll have more focus, they'll get better grades, they'll have more energy to play sports And there'll be much more masculine if they use these things. But using nicotine in children is very dangerous because some of you aren't even 25 yet, probably, and brains aren't even fully developed, but yet you've done such a great job. You're in medical school. What we see is that some of these kids who are starting at 8, 9 years old Are using so much nicotine that it's affecting their brain development And then they're winding up on other gateway drugs.

They have a hard time if they become addicted with obviously their mood, their attention, their impulse control And the overall point is hammer home not safe for anyone, but certainly not safe for our young population. Who cares? Okay, hookahs are not safe either and the main thing that I want you to know is that one session is you're inhaling about five packs of cigarettes.

That's 100 cigarettes. A pack is 20 cigarettes. So in one hookah session, you are breathing in 100 cigarettes.

Avoid hookah parties So there's a lot of benefits to quitting smoking. Some of them, some of the benefits happen within minutes, and then others take years, many years, it takes up to 15 years for you to go back to a safe cardiovascular baseline So what's awesome is that we're seeing that less kids are smoking and less adults are smoking Tobacco cigarettes But that means that more are doing e-cigarettes or vaping. So What we're seeing that in the United States, even though that there is a decline in the overall world that it's still a major problem.

So one in 5 adults are still using tobacco products worldwide, more in men And kids all things are declining except for the pouches. There was a sort of seed plateau. But in kids recently we're seeing that vaping is coming down and cigarette use is coming down. So we're doing something right Recently, but it's not enough. So Smoking cessation is a huge part of what you need to learn how to do as providers to help your patients Because a lot of patients want to quit, right? We even heard from young adults that they wanted to stop vaping, but they just couldn't. Half of your patients will be attempting to quit It takes on average, 30 times to successfully stop smoking And only 4% to 7% of your patients will be successful in stopping smoking. And the reason why is it's very Complex as to why people get addicted.

Sometimes it's socially... for social reasons, psychological reasons, biological reasons, or all of the above. And you have to realize that it takes a multidisciplinary approach to successfully get your patients to stop smoking.

So it's usually done with behavioral counseling of your patients and supporting them that way, but then also with medications. And then there are alternative things that you can talk to your patients about as well, such as acupuncture, hypnosis, or other type of natural products. So there's only a certain amount of FDA-approved products to help with smoking cessation and the mainstay is your nicotine replacement therapy.

And that has to do with your literally giving them the nicotine that they're addicted to, to try to get them off of the tobacco products that they're using. And the most common way of doing it is giving them a patch that they put on, which is a baseline amount of nicotine that they have in their system, and then when they have cravings, they use something else. So that something else could be gum or a lozenge. Then you have pills that they can take.

There's one called Zyban and another called Chantix, and they work essentially to reduce the withdrawal symptoms from stopping smoking.

The Chantix has better success rates than the Zyban. And they all have the do's and the don'ts, which you'll learn as you go through your training, but these are very successful to help your patients stop. There's a new drug that the FDA is almost approving.

It binds to the same receptors as Chantix. So it has very similar properties to reduce those symptoms of withdrawal and cravings. It's still under review.

But it's very promising. So they think that it might be out on the market this coming year in 2027, which is really exciting because we haven't had a new smoking cessation drug in a long time. Again, patients are going to try and fail at least 30 times, which you have to support them every time that they fail, saying it's okay. The majority of people fail. Let's try again. That didn't work. Let's try this. Let's maybe go for, you know, add in some acupuncture or hypnosis to the other things we're doing as well. At some point, if you throw enough things against the wall and they're willing to keep trying, at some point it does stick, and then it's it's game-changing, life-changing for them. So again, the counseling part of it, it's extremely, extremely effective. So even if you sit and talk with them for just a few minutes, it really makes a lasting impact on them. All right, so this is a huge new following that I have in my pulmonary clinic. Is patients using cannabis and cannabis-associated lung disorders that are coming in in the spectrum of things is cannabis smoking safer than cigarette smoking?

Yes. Is it safer than smoking than vaping? Yes, most of the time. Okay, so I think it's important for you to know the risks. And maybe some benefits to using cannabis. So, cannabis is, it's a plant-derived psychoactive substance.

It contains that active chemical compounds called cannabinoids. The most notable ones are THC, that's the one that gives patients the high that they're craving. And then CBD, which does some of the pain modulation of what that some patients want to go after that arm, the CBD arm. So these things will regulate your mood, your appetite, your memory, and pain. They can be consumed in different ways, so they can smoke it, they can vape it, they can eat it, they can dab it. And each one has different safety profiles.

Out of all of them, if they're trying the safest with cannabis is to be the edible forms. But it often doesn't achieve what the person's looking for. Most people want to inhale it to get the most fastest onset of effect.

And depending on how they inhale it and which way, it could either take about 15 minutes for them to get their desired effect, or it could happen really within almost instantaneously minutes if they do it another way. So these are the different things that you might see. So we have, and you guys might be all well educated in this, but maybe some of you have never actually seen this stuff before, which is why I put it on the slides. So this thing right here is what we

call a joint That's literally plain rolling paper with cannabis flower, marijuana flour, the plant.

This This right here has marijuana and tobacco in it. So this is a bigger no-no, right? But a lot of patients do that because they like the mix of getting calm, the high of the cannabis, but then they get a little too... a little too high, and they like the tobacco speed, so it sort of, like, cancels some of the side effects out. This is a spliff. Then we have over here, this is what patients would call blunts that they're smoking blunts.

So this is a cigar paper roll that has marijuana and tobacco in it. So out of the safest one, this one would be the safest one.

It's not still not healthy, but educating your patients about the choices they make, I think, is important. These are the different ways that you could consume it.

So if a patient says, I smoke it in a bowl, these are bowls. Usually they're glass And they put in the flower marijuana, and they heat it up under the part that holds the marijuana, and they inhale the smoke. This is a bong.

So this is they'd put the marijuana over here. There's water in here. And they would heat it And then smoke it from up here. So A bong is left safe than a pipe because bongs have to be cleaned out. So if they don't readily clean out this reservoir, there can be mold in gosh knows what else that grows inside of there that they are then actively breathing into their lungs. These are some edibles.

So there's gummies that they can put it inside the flower inside and make brownies. But there's all different types of now like sweets and treats that patients can go and buy And what's important is if you have adult patients using these things who have young children, you need to actually educate them that yes, they're doing this for their enjoyment. But if their child comes along and thinks that these are gummy bears, their child will most likely wind up in the emergency room with THC overdose Marijuana is increasing big time in the United States, and I'm sure you guys have all seen that personally. I find it crazy that 15% of all of our population are actively using marijuana About 22% of those being people 12 years old and up There is a increase in the older population.

So the trend more recently over the years has been that more adults 65 and older are using it. More women are consuming cannabis in the United States.

The way that they like to consume it the most is by edibles In the female population And they think that part of the reason why cannabis is on the rise is because They're using less alcohol.

Alcohol consumption in the United States is really, really low at this point in comparison to other years. So something that you should look at is that Cannabis sales in the United States actually really helps boost the economy, but it's actually not lucrative I found out for people who own dispensaries.

So the government is really benefiting from it, but not like private business owners. So the good news maybe is that the marijuana use in teens has been stable since over the last 4 to 5 years.

And the way that teens prefer to use marijuana is in the vapor pen. So teens don't like to smoke joints Use bongs or use pipes. They like to use the vapor pen, and that's most likely because Their generation is used to vaping nicotine products, so they're just used to this type of delivery. So some people might ask, your patients might say, well, is it better for me to smoke my marijuana or is it better for me to vape my marijuana? And you literally have to give them educated information about this. So you know, I... there's a table sort of just going over the different Pros and cons or risks and benefits, whatever you want to say, of how they inhale their marijuana, and What I would say the biggest

thing to think about is that when they vape it The concentration is much more potent, so they're getting way more of the psychoactive drug than those who are using, like, joints or bowls or something like that. Also The difference in lung health is that in those who are smoking it, you're more likely to develop to develop like a chronic cough Or even you can develop obstructive airways disease versus those who vape marijuana.

Those are the ones that we've seen who've had the acute lung injury, like an ARDS pattern, and they can also get ongoing chronic lung injury that can happen from ongoing chemical burns because it's just so hot and they're breathing it in much deeper into the lung units than when they smoke the flour, because the smoke there sort of They don't take as big of an inhale. So it's just a difference in the pathology that can happen depending on how they smoke it. So I would say that The vaping marijuana is not as safe as smoking the flour of marijuana. So some patients will take it a step further and be like, well, no, no, I dab it. You know, it's not the same as vaping. Like, I dab it so that it's even way safer. I'm gonna get rid of, like, all of those potential chemicals, I'm gonna get rid of you know, anything to do with the with the flower plant, because that's going to be safer for my lungs. But It's definitely not healthier than vaping.

It's actually been shown to be The most likely to cause lung damage and the most likely to wind up having cannabinoid use disorder. So DAB comes in all different, like, shapes and sizes and consistencies, but essentially it's a very, very, very concentrated form of cannabis It is made to be either solventless or solvent-based.

And solvent Less means that they don't have other things as they create them, such as butane, propane, or alcohol during extraction And then other ones could have some solvent residual in there, so you can see that breathing those things in is not healthy for your body, right? So You need to be able to talk to your patients about, well, if you're dabbing, well, what product are you dabbing? And do you know that using that type is riskier than using this type? So what you need to know is that it's way more potent And that You likely will become, like I said, a higher risk of cannabis use disorder than those who vape and smoke flour alone. It takes a very, like, complicated device to dab as well. So again, it's another thing about they have to clean it out. It's an investment that they go into because they're seeking a much faster high, and they think it's healthier Okay, so really try to tell your patients to move away from that if they're doing that There are a lot of health risks associated with chronic use of marijuana.

And that is patients are now using it and then driving or going to work. I have a number of employees that come and see me throughout the day for just their routine checkup, and I can tell that they've been using marijuana There's no regulation on that yet. So that's why I would say that it's pretty dangerous. And in the long run, it can cause worsening mental health.

It can cause chronic long-term respiratory problems. You're more likely to abuse other drugs, and there's a risk of developing cancer. So With this, we're going to take, like, a few min break maybe like 5 min or so, and we will come back to talk about prevention. Okay We'll get through the final stretch We're going to shift gears.

And now talk about extremely important things For you to know to talk to your patients and you know when you should be talking to your patients about immunizations. Okay. So there is a handful of immunizations which are extremely important for pulmonary prevention, and that's our flu vaccine, COVID vaccines, RSV vaccine, pertussis, and pneumococcus.

Those are the main ones that you want to make sure your patients are covered for to prevent morbidity and mortality. So some of the most common things that we can talk to our hands about, especially during flu season, is that they frequently should be cleaning their hands, washing their hands And I guess all

ages of your population, you should try to teach them how to, like, cough or sneeze into their elbow So that they're not just like viewing it all over everybody and transmitting their disease to everyone. We should really tell people or encourage people that if they're sick, especially with a fever, that they should stay home.

I think we are really bad at that as medical professionals because we have so much responsibility And then we have this crazy, like, need to just never let anybody down, but I'm telling you, as students and as future physicians, if you're sick and you're seeing your patients who you put them at risk, then that's really not ideal, even though you don't intend to To harm them, you really could, especially if you're in your acute febrile phase You want to tell them to get their To clean their surfaces.

So Sometimes they might not understand that doorknobs are highly infectious areas or and that's doorknobs even like To out the outside of the house, the bathroom, things like that. You want to just have them wipe things down. Try for them to not be touching their face so much. That's very much in kids And then wearing the mask, there's depending, you know, for us if we have some type of communicable disease and we come back to work, there are guidelines to guide us to say, well, for this many days you should, if you just had COVID, you should be wearing a mask And that's to protect other people So Flu still kills a lot of people and it's That's really preventable, so you want to make sure that you tell your at-risk populations to get a flu shot.

And it's really important more on like the extremes of ages is where we see the most amount of diseases, but those who have underlying comorbid medical conditions, you should also really encourage them to get their vaccines. So you're probably having your micro classes or whatever about the different types of flu that are out there, and that A and B are the most important for us in humans and that there's the HA and NA surface antigens and that you can have drifts and shifts depending on what happens Genetically, so if you have a drift, there's little small changes that the flu virus does to try to evade our immune system every year, and then a shift is when it's a really huge one, and that's when we have something bad Like we saw with Edward over here, right? Okay, so essentially every single person should get a flu vaccine, so it is approved for anyone 6 months old and over, and there are different types of flu vaccines on the market that you can talk to your patients about But the most common ones that we're used to are the standard needle ones, but some people are freaked out by needles, so you can tell them that they're a needle-free vaccines. And then in your older populations, you give them the higher dose one because they're less likely to respond, make an immune response to the regular one. So that's why you just sort of give them more of the antigen So that they can actually boost their immune response. So the flu vaccine, though, is the efficacy is not that great. So when we are pushing, pushing, pushing for the flu vaccine, we would hope that we'd get it right. But it's so hard for our scientists to get it right because this thing is so smart and it's always evading us so that we create the vaccine and put it in production, and then the thing changes, right? So But no matter what, if we didn't vaccinate, there'd be a ton of more deaths COVID-19.

All right. This, I think You know, you would certainly have all been exposed to affected by and have caught, right?

So It's no longer a super deadly disease unless you have the comorbidities. Thank goodness, right? But essentially The parts of the body that it affects are due to the parts of the body that have the ACE2 receptor.

And there are people who had the original strain, the 2020 strain, who are still affected by COVID. And I met a new patient in the office today Long COVID syndrome, and she is miserable.

So, thankfully, the thing has mutated into less virulent strains that are now not causing long, long, long-term sequelae. So what you should know is that

People, it's recommended for people to still be vaccinated against COVID even though it's not as deadly as the flu, but if we stop vaccinating, you know, it could get bad again So there's three different types of vaccine on the market, and they are approved again for six months and up And there are people who can get boosted for it, and those who you should suggest getting boosted are the ones who have the comorbidities, right?

Your diabetics, your end stage renal disease, your extreme, you know, your elderly population, those who are living in nursing homes Very commonly, people are like I just had COVID. Do I need a booster the thought process is, is that it's okay for them to wait at least 3 months for their booster. So a ton of people have been vaccinated. But there's still a lot of weary people in terms of vaccination. I think you're going to Continue to come across that, especially depending on whoever, you know, is in, you know, the political leaders of the United States.

So just in my lifetime, in the 6 years of COVID, we've shifted from everybody must 4 million COVID vaccines to nobody should have a COVID vaccine, but as medical providers, we still need to know what the literature supports and boosters are still scientifically and medically supported, so you should be talking to your patients about that. Because boosters work. Okay, RSV.

This one is one of the newer vaccines that they are still adjusting and making guidelines for. And You should really know about RSV because it can affect everybody.

I'm sure some of you guys had a cold that was RSV this past winter. It can significantly impact, again, the extremes of age So for us, we've had this infection probably a number of times, and every time we get it, our body is able to mount an immune response so that as the severity decreases and decreases. So Children who are quite young will be most oftentimes hospitalized with a respiratory problem.

A lot of them wind up needing to go on ventilators, and it's most common in the very, very young kids. And then in adults Those who are 50 years old who have underlying medical conditions or in the elderly, those are the ones who are going to have the biggest trouble with this virus. And what you should really take away is that the mortality rate can be quite high. So in a person who's 50, who's at risk, and they get hospitalized, their chance of dying from RSV related infection is Upwards of 8%.

That's huge. So there's different presenting symptoms in kids as you go through your different rotations, you'll see how babies present to toddlers.

They're oftentimes very noisy breathing and nostrils are flaring. They could become cyanotic and they need to get to the hospital Whereas we might be getting, oh, we feel a little fatigued, run down, we might be coughing, we might have a fever, some, you know, upper respiratory stuff There are three vaccines available.

And essentially what you should know is that They happen in pregnant women.

They're given to babies, and they're given to adults 50 years and older. So the recommendation is Anyone 50 years with comorbidities Should receive a RSV vaccine, but everyone over 75, the recommendation is for an RSV vaccine. And then there's only one of the three vaccines, which is approved in pregnant women. And this is very... this is minutiae stuff, and this is never something that would be on a test question. It's just for, I would think, interest... be interesting for you to know as you go through your GYN rotation, for instance, that if that you're going to be looking out for your Pregnant patient who is, you know, between 32 and 36 weeks of pregnancy, you want to be talking to them about getting their Rsv vaccine to protect their baby. So some of the comorbidities are, again, those who have if they have an underlying COPD, emphysema, if they have heart disease, if they have diabetes, if they're

immunocompromised because they have a malignancy or something, if they're on end-stage renal disease, or if they have any type of underlying neurologic problem. Those are the ones that you should be Thinking about between the ages of 50 and 74. And at this point, it's not like an annual vaccine. It's just a one time vaccine. And if you vaccinate them, you're really decreasing their morbidity and mortality.

It reduces the risk of them getting hospitalized by 75%. So that's pretty big. This is just a slide to help you guys. If you're interested in learning more about it as you start going through your rotations, it sort of simplifies it for you. The whooping cough is another one that we should be vaccinating. So all pregnant women get it.

It's recommended now for spouses and close family members of pregnant patients to get it To prevent the whooping cough. So the whooping cough is something that comes in stages. So the first stage is you're sick, like you have a cough, a runny nose. The second stage is that you could be like in these coughing fits for months And it's like, literally fits and spasms where you're vomiting sometimes from it.

Your Like whooping, so you can't catch your breath, so you're coughing so much. Ow, ow, ow, ow, out, and then finally that whoop sound is because you finally get to take a breath in, and that's what... that's the classic whoop And then after a couple of months, you finally start gradually starting to improve.

So you might have even had this yourself and have not Known that it was the whooping cough, but you've just coughed for months and months and months.

It's very easily treated if we diagnose it, but it's certainly very preventable with the vaccine Okay, pneumococcal disease.

So, another one that you want to vaccinate so that patients don't wind up getting the bacteria in their blood, they don't get pneumonia, they don't get meningitis, And also it decreases ear infections, especially in little kids. So underlying diseases, chronic lung diseases, you definitely want to be vaccinating your patients. So there's all different ones on the market. And essentially, there are charts and guidelines that you can follow, but if they've had it or never had it. And again, these are not things that are memorized. These are things that you'll learn through your rotations. This is an introduction to the different types of vaccines that are out there. You might have a patient on a rotation that says Should I get Prevnar 13? And then you're going to look at your attending and be like, what is Prevnar 13? And you'll talk about it and read about it, because there's the 13, there's the 20, there's the 23. And there's different indications that get followed. So We'll come back to our cases.

So our first case, our young person, we're going to say that they should have the flu vaccine and a up-to-date COVID vaccine. Our second case, they should have, these are the vaccines that That they should be thinking about, oh, and I should update this for next year to 50 because they changed the guideline over the year. Okay, so we went over vaccines. So now there are some other topics and we're almost done with the talk, but this is a big one.

This is latent tuberculous infection, and you need to know that this is different than tuberculosis. So TBI is latent TB.

So TBI means that you were exposed to TB and that your immune system was able to Put it at bay that you did not wind up developing active TB, but it's still in your... in your body, and if something happens, it can reactivate, and you can develop active TB. And that's the point of what's so important about TBI Is because as providers, if we diagnose it and treat it, you're going to prevent your patient from developing tuberculosis.

Not only will they be sick, but then they can make other people sick. So you

have to think about, well, who are we gonna test for it, and who are we going to treat? So, for people with TB infection, no risk factors, and you need to know a little bit about the history of it. So, if they have TB, no risks, they don't get treated. The risk is about 5% in the first 2 years after infection, and about 10% over a lifetime of developing active TBA. About half of those people who develop TB will do so within the first 2 years of infection. For people with untreated TB infection and untreated HIV, the risk is about 7-10% per year. TBI has been decreasing steadily over the years because we've been picking it up better and we pick it up better by screening people.

So we've been asking the questions of certain populations that we think need to be screened. And then they can be tested. So this is very reassuring, however COVID happened. And during the COVID pandemic, all of the work that we were doing to be screening and watching people take directly observed therapy, all of those resources were shut down and a lot of those healthcare workers we're switched to being the COVID chasers, essentially. You have COVID and then telling the people around them that they have COVID and checking on them. So it's sort of fell apart a little bit, and it started to increase again from 2022 on. But we're doing better, guys.

So it's now started to come down in 2025 because we have more of a handle on prevention and we're able to put our resources into this important preventable disease. So that's good news. So how do we diagnose it?

So there are two types of tests that we can use to detect the bacteria in the body. It doesn't tell you if you have active tuberculosis or TBI, though. It just detects the bacteria. So most of you guys have probably had a PPD, but they've actually been shifting to do a blood test instead. So PPD is a skin test, whereas the other one is a blood test but not at a routine place. You have to be able to be at a lab that can specifically run the Quantiferon test, which is what it's called. So we want to screen those who are at high enough risk that they might reactivate and then develop active TB.

So those are the patients who have HIV because we already saw that they're more likely to become positive than the general population without risks up to 10%.

So HIV, if they're IV drug users, if they have end-stage renal disease, diabetes, silicosis is a lung disease, if they're on immunosuppressants. If they have cancers, if they are malnourished, and if they've had a gastrectomy, those are your population of patients that you'd consider screening. So when you screen people, you break it down into risk. So those at high risk of acquiring it are those who are living in the same house or have close contact with them. If you work in a mycobacteriology lab, you're exposed to it frequently, so you're considered high risk. Immigrants from countries with high rates of tuberculosis are at the highest risk, I would say, in the United States right now. Those are the ones that have the highest rates of positivity. If you are living in... if you're a resident or you're a staff for a homeless shelter or even a correctional facility. And then the populations that we just discussed. So we'll go on to, if we want to use the tuberculin skin test. It is a delayed type hypersensitivity response that gets mediated by T lymphocytes.

And in those patients who had exposure, it will cause induration, a swelling at the injection site within 48 to 72 hours. So it's not considered that it got red. But that it actually gets hard and raised.

It has it feels like the consistency of a pencil eraser. And that's that's considered what the end duration is. And you have to measure the part of the firm pencil eraser, and depending on how big that is and the risk factor for the patient will determine if they should receive medication. So there's different cutoffs for different populations.

So if you have a 5 millimeter or greater cutoff you're considered positive and would be a candidate for treatment. If you are a person who has HIV, if you are someone who's in close contact with somebody, if you have fibrotic changes on

your X-ray If you have an organ transplant, or if you're an immunocompromised patient. So the lower millimeter and the higher risk patient, you would get treated. If your 10 millimeters or greater, then we would treat you if you are born in countries that are high prevalence, such as Mexico, the Philippines, Vietnam, India, China Haiti and Guatemala.

If you're someone who misuses drugs, if you depending on what your occupation is, if you're a mycobacterial mycobacteriology lab worker Those who are high-risk medical conditions, if you're less than 5 years old, or if you're a kid who's exposed to adults. And People who are, if you test greater than 15 Would only be for those who you screened and they don't really have any risk factors, but they're interrater really large, then you would treat them. So That's sort of how it's broken down. So you can get false negatives and false positives for your tuberculin skin test. False negatives could be that they're immune-suppressed so that they're not able to mount the immune response to make that in duration. Or if they were exposed to recently and the body hasn't had time to mount to it, or if People don't know how to interpret the test. a false positive could mean that maybe they have mycobacteria, but they don't have tuberculous mycobacteria, so something called, like, MAI right Or in patients who have had the BCG vaccine, they could still mount a tuberculin skin test response. So that's why, because of these sometimes people don't know how to interpret the testing. You know, there's There's not... there's things that are subjective about it That makes it where we want to do the blood test and also the fact that BCG vaccines and other types of mycobacteria can affect the interpretation. They shifted a lot of the times to doing what's called a quantiferon And the interferon gamma release assay.

And the reason why they do these is because it's objective. It's either yes, maybe no, kind of a thing And it's one and done. So with the tuberculin skin test, you put it in, and then you have to wait the 48 to 72 hours for them to come back for you to interpret it. Sometimes patients might get lost in there, and then you'll never have your answer. So you do these tests As a positive negative and an indeterminate or a maybe. If you have an indeterminate patient, you just ask them to come back again in about 2 months and repeat the test then. So there's different regimens that you can use.

INH has fallen out of favor due to its toxicity. So it's very common for patients to do their rifampin Regimen, which is essentially you take rifampin daily for 4 months, so I put the majority of my patients on this regimen because it's easy And it's safer. So other things that you have to think about for prevention, pulmonary prevention are preventing blood clots So The most common presentations of venous thrombosis are DVTs, so deep venous thrombosis of the lower extremities, and then pulmonary embolisms. So there are different reasons why you can develop these things. There are Blood reasons like inherited risk factors listed there, and then there are acquired risk factors.

So I would say the most common risk factors that I see are in those patients who have an underlying type of malignancy, and in those patients who who are hospitalized and they are not given DVT prophylaxis But there are a number of reasons listed here why they could potentially develop thrombosis. So a pulmonary embolism is a direct complication of a deep venous thrombosis.

So This here is your pulmonary artery.

So this is your ascending aorta. That's your descending aorta, and this is your pulmonary artery, which goes into the right main and the left main. This dark stuff here Is all blood clot, so this is essentially blocking off Both pulmonary emboli and this thing here, both sorry, both pulmonary main pulmonary arteries.

And this thing here is what we call a saddle pulmonary embolism, and the saddle just means that as if it was sitting on top of The bifurcification and saddling both sides. This right here, when you see this, this patient has the potential of death Right? So there are certain things that you would look at For how risky they are and how you need to manage them. But what's important for you guys to

know is that Blood clots are distal and they come proximal So we have to prevent them, and that's your job as the providers when your patients get admitted to the hospital, you have to think who should get DVT prophylaxis and who shouldn't. And there are different types of reasons why they should or should not. So if somebody Comes in and they're healthy, and they're having an ambulatory procedure and going home the same day. They technically would not get DVT prophylaxis, but if they're coming in because they have COPD and pneumonia That patient needs DVT prophylaxis.

If they're having an orthopedic procedure, that's a huge high risk. They need DVT prophylaxis. So if you don't prevent it then they can develop it, and then they could develop a fatal PE. And that's what you want to prevent. So the slide here is just trying to say that 50% of all reasons our healthcare associated, so we have to do a very good job of protecting our patients and preventing them. So how do we do that? Well, you get your patients up and moving as fast as possible. I'm amazed by orthopedic what's happened in orthopedics lately, where people can have joints replaced, and they're like, get up, run that marathon, get up those stairs, and It seems almost like unfathomable to me if I have to experience that, but it's done to prevent morbidity and mortality because it gets them up and moving Other things that you could do are put on these pneumatic compression devices, which are essentially like a blood pressure cuff that gets blown up and down on the legs, and it simulates the muscles contracting and pushes the blood So that helps prevent blood clots. Or you can give Heparin Or you can give low molecular weight heparin.

So they could be drips where it's a constant infusion, or they could be shots, or there could be pills. And all of these things help prevent blood clots. So sometimes patients only, they don't need any Medication Other times they need medication only when they're in the hospital, and then other times they need medication when they leave the hospital, sometimes even over a month later for certain orthopedic procedures, so that they don't get blood clots can also be seen in those who have, like, abdominal or pelvic cancer surgeries.

Those patients are also given extended DVT prophylaxis. So in all of our patients, if we have to look at their risk factors to consider, do they need a PPD or a quantiferon test to rule to look for TBI And certainly if they are on immune-modulating medications That's it I appreciate your attention, guys. Questions, concerns Suggestions for next year's class? Okay. Thank you again.